

P100-001: Exact-prefix sequence extrapolation

Scenario and question

The OEIS integer snapshot is sampled by a fixed accession hash. Each eligible sequence supplies 16 exact prefix terms and one five-step-ahead target. Affine-equivalent observed prefixes are linked before splitting. Sequence names and source descriptions are excluded from prediction.

When can compact exact arithmetic hypotheses extrapolate a prefix, and where do they fail?

Experimental contract

Only first 16 terms and horizon 5 supplied; names, OEIS IDs and published formulas are forbidden predictors. Sixteen observed prefix terms per sequence, including confirmation sequences; no suffix term used in rules.

The campaign completed 7 substantive hypotheses. Native bytes and exact sample anchors are preserved. Model selection uses whole-group development error and a 1% simplicity tolerance; selection and stopping were frozen before the separate no-fitting confirmation process. All outcomes are now exposed for future users.

Main result and interpretation

Seven exact-family investigations test finite differences, recurrences, residue classes, rational ratios, polynomial forcing, a cascade and an internal falsification gate. They do not beat the constrained learned comparator on the primary signed-asinh endpoint. Exact prefix agreement is compatible with many incompatible continuations; it is not a proof of an integer-sequence law.

The reference uses a Gaussian radial-basis residual of all 16 prefix terms in signed-asinh coordinates; the baseline is the last observed transformed term. Exact recognizers and their failed continuations remain separate scientific evidence.

The reference predicts the transformed twenty-first term from the first sixteen terms:

$$\hat{y} = \operatorname{asinh}(a_{16}) + \beta_0 + \sum_{k=1}^{24} \beta_k \exp(-\|z - c_k\|^2/8).$$

Here $z_j = (\operatorname{asinh}(a_j) - \mu_j)/s_j$, $j = 1, \dots, 16$. The development-fitted means μ_j , scales s_j , centers c_k and all 25 coefficients are recorded in rules.json. Ridge penalty 0.3 is fixed. No sequence name, accession or observed suffix enters this prediction.

Parameters: 25 frozen coefficients, feature scaling and centers are listed in rules.json; no hidden fitted state. Coefficient ordering follows run.py and rules.json.

Performance and adverse evidence

Model	Development MAE	Confirmation MAE
rbf	2.02965	2.23335
persist	3.08003	3.35643

MAE is averaged within each complete group and then equally over the 474 reserved groups (479 observations). Errors are in asinh(integer term). Per-group RMSE, bias and failures are available in evidence/by_group.csv; small or dependent group counts do not support population confidence claims.

The lowest retrospective confirmation error belongs to rbf. It did not change the frozen selection. Seven exact-prefix families remain worse than the learned comparator on the declared transformed endpoint; a formula that fits the prefix is not confirmed as an infinite rule.

Value, limits and evaluation

A radial-basis residual predictor is retained as the numerical reference. This provides a challenging open extrapolation task and preserves exact mathematical counterexamples, but it does not establish new combinatorial principles. A finite prefix cannot uniquely identify an infinite sequence. Related OEIS families beyond the linked affine prefixes can remain dependent. Very large integers are scored in signed-asinh coordinates, not as a percentage of exact integer matches.

Alternative agents can submit any predictor under this task's information budget. The shared evaluator reports task-unit errors, whole-group and worst-group results, coverage, abstention, optional intervals and meaningful event diagnostics. A different endpoint requires an independently reviewed new-task contract. Numerical success alone does not certify mechanism, novelty or deployed impact.

Reproducibility and source

Run `python run.py` for the frozen reference replay. `rules.json` contains every fitted value; `task_spec.json` defines units and access; `evidence/` contains compact predictions and numerical evidence. Research history preserves all hypotheses, failed candidates, technical corrections and the first confirmation receipt.

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